

# 01\_overview\_er

## 1. Overview & Layered Model

Helios transforms raw GA4 e-commerce telemetry into a governed, Kimball-style star schema that an autonomous multi-agent system can query *without ever authoring SQL by hand*. The data model is the foundation under everything else: the semantic layer ( `semantic_layer.yaml` , 47 metrics) resolves every metric against a small set of mart grains, and the agents compose those governed metrics rather than touching the warehouse directly. This document is the canonical, production-grade specification of every table in that pipeline — its grain, primary key, foreign keys, owner/steward, and, most importantly, *why it exists*.

### 1.1 The layered DAG

Transformations flow through five layers, each with a single, non-overlapping responsibility. Building one logical step per layer is what keeps the warehouse cheap to query, the lineage legible, and downstream models insulated from upstream change.



#### Why each layer exists.

- **Raw source** ( `src_ga4.events` ) is the immutable source of truth owned by Google's GA4 export. Its rows are *events* with deeply nested `event_params` , `items[]` , and `ecommerce` structures. We read it exclusively through `source()` so that exactly one model touches the export schema; nothing downstream depends on the raw layout.
- **Staging** is the 1:1 typed/renamed access layer. It surfaces the scalar columns analysts actually need ( `session_key` , `ga_session_id` , `page_location` , session-scoped source/medium, `device.*` , `geo.*` , `transaction_id` , `purchase_revenue_in_usd` ) and flattens the nested `event_params` array exactly once so no downstream model ever re- `UNNEST` s. Staging isolates the entire pipeline from raw schema drift; materialized as cheap, always-fresh views.
- **Intermediate** holds the business logic that the raw export omits. GA4 ships no session row, so `int_ga4__sessionized` reconstructs one by grouping events on ( `user_pseudo_id` , `ga_session_id` ) and `int_ga4__funnel_steps` computes the six monotonic `reached_*` funnel flags. These two are the *keystones*: get them wrong and every downstream number is silently wrong. They are ephemeral/view — never exposed to BI.
- **Marts** are the conformed star schema: wide, denormalized fact tables surrounded by conformed dimensions, materialized as tables (incremental on the large facts) so the autonomous run hits a fixed, predictable BigQuery byte budget.
- **Semantic** is a thin governed view over the marts. It is the *only* path to SQL — `semantic-mcp.build_query` maps a metric name plus dimensions to mart columns, so an agent that asks for `revenue by channel_group` never sees a `JOIN` or a `SUM` .

### 1.2 The two grains-of-record

Helios has exactly **two entities of record**, each anchored to its own fact grain:

- 1. **The session** — one analytical visit, keyed by `session_key = TO_HEX(MD5(CONCAT(user_pseudo_id, '-', CAST(ga_session_id AS STRING))))` . The session is split across two co-grained facts: `fct_sessions` carries engagement and the wide session descriptive dimensions, while `fct_funnel` carries the monotonic `reached_*` flags plus deduped `session_revenue` . `fct_funnel` is the **primary** session grain the semantic layer queries — funnel, conversion, and revenue-per-session metrics all resolve there. The two join 1:1 on `session_key` ; the funnel *extends* the session.
- 2. **The transaction** — one completed order, keyed by `order_key = transaction_id` . `fct_orders` is the deduped order header (gross/net revenue, refund, shipping, tax, quantity) and the entity of record for all order-grain financials. Its line items explode into `fct_order_items` .

A third, derived analytical entity — the **user** (`user_pseudo_id`) — is conformed in `dim_users` but is deliberately *cookie-grain*: `user_id` is almost always NULL in this dataset, so there is no cross-device stitching. One human on a phone and a laptop is two users. This is carried as an honesty caveat on every user-grain metric (ARPU, new/returning users) — they are per-cookie approximations, never true person counts.

1.3 Conformed dimensions

Four dimensions conform across the facts so the Decompose agent can pivot any metric across any slice without re-deriving the slice:

- `dim_date` — the date spine (2020-11-01..2021-01-31) with week/month/quarter/day-of-week/is\_weekend for time-grain rollups.
- `dim_users` — the user entity-of-record (cookie-grain) with first-touch attribution and lifetime rollups.
- `dim_channels` — the 10 GA4 default channel groups produced by the single `channel_group_case()` macro, plus `is_paid / is_organic / channel_order` .
- `dim_items` — the product catalog (item name, brand, category 1..5, current price).

Because these dimensions are *conformed* (one shared definition, joined by surrogate key), a finding sliced by `device_category` in the funnel and a finding sliced by `device_category` in revenue use the same dimension values — the precondition for honest mix-vs-rate decomposition.

1.4 Mart → semantic grain mapping

The semantic layer resolves every one of its 47 metrics against exactly **five base grains**. The mapping is explicit and 1:1:

| Semantic grain               | Backing mart                 | Entity     | Metrics resolved here (examples)                                                                                                                                                                                                                                     |
|------------------------------|------------------------------|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>fct_funnel</code>      | <code>fct_funnel</code>      | session    | <code>sessions</code> , <code>users</code> , <code>new_users</code> , <code>returning_users</code> , <code>session_conversion_rate</code> , <code>view_to_cart_rate</code> , <code>revenue</code> , <code>revenue_per_session</code> , <code>revenue_per_user</code> |
| <code>fct_sessions</code>    | <code>fct_sessions</code>    | session    | <code>engaged_sessions</code> , <code>engagement_rate</code>                                                                                                                                                                                                         |
| <code>fct_orders</code>      | <code>fct_orders</code>      | order      | <code>orders</code> , <code>gross_revenue</code> , <code>net_revenue</code> , <code>aov</code> , <code>items_per_transaction</code>                                                                                                                                  |
| <code>fct_order_items</code> | <code>fct_order_items</code> | order_item | product-level revenue, attach rate, units                                                                                                                                                                                                                            |
| <code>fct_cohorts</code>     | <code>fct_cohorts</code>     | cohort     | <code>day_1_retention</code> , <code>day_7_retention</code> , <code>day_30_retention</code>                                                                                                                                                                          |

The remaining marts — `dim_*` , `fct_daily_funnel` , `fct_funnel_by_dim` — are conformed dimensions or pre-aggregated rollups the agents read *indirectly*. `fct_daily_funnel` and `fct_funnel_by_dim` are additive aggregations of `fct_funnel` that feed the Monitor (anomaly detection), Decompose (mix-shift), and the eval injector; they store additive counts only, and all *rates* are computed in the semantic layer from those counts so they re-aggregate correctly across any slice.

1.5 The Kimball / wide-fact philosophy

Helios follows dimensional modelling deliberately. Each fact table declares exactly one grain; conformed dimensions are shared by surrogate key; and the facts are **wide** — descriptive dimensions ( `device_category` , `country` , `channel_group` , `is_new_user` , `landing_page` ) are denormalized directly onto `fct_funnel` / `fct_sessions` / `fct_orders` . The semantic layer therefore slices a metric by simply `GROUP BY` -ing a column on the same fact, with no join at query time. That is what lets a Sonnet-class agent request " `revenue by channel_group and device_category` for last week" and get governed, reconciled SQL the agent never wrote: the breadth lives in the mart, not in ad-hoc query construction. Wide facts trade a little storage and ETL discipline for *zero* analyst-authored joins, which is precisely the property the grounding principle ("the LLM never authors raw SQL") requires.

2. Entity Relationship Diagram

2.1 Mermaid ER diagram

```

erDiagram
    dim_date {
        DATE date_key PK
        INT64 week
        INT64 month
        INT64 quarter
        STRING day_of_week
        BOOL is_weekend
    }
    dim_users {
        STRING user_key PK "= user_pseudo_id"
        STRING first_channel_key FK
        TIMESTAMP first_touch_ts
        INT64 total_sessions
        FLOAT64 total_revenue
        BOOL is_purchaser
        BOOL is_new_user
    }
    dim_channels {
        STRING channel_key PK
        STRING channel_group
        BOOL is_paid
        BOOL is_organic
        INT64 channel_order
    }
    dim_items {
        STRING item_key PK "= item_id"
        STRING item_name
        STRING item_brand
        STRING item_category
        FLOAT64 current_price_in_usd
    }
    fct_sessions {
        STRING session_key PK
        STRING user_pseudo_id FK
        DATE date_key FK
        STRING channel_key FK
        BOOL engaged_session
        STRING device_category
        STRING country
    }
    fct_funnel {
        STRING session_key PK "FK → fct_sessions"
        STRING user_pseudo_id FK
        DATE date_key FK
        STRING channel_key FK
        BOOL reached_view_item
        BOOL reached_add_to_cart
        BOOL reached_purchase
        FLOAT64 session_revenue
    }
    fct_orders {
        STRING order_key PK "= transaction_id"
        STRING session_key FK
        STRING user_pseudo_id FK
        DATE date_key FK
        STRING channel_key FK
        FLOAT64 gross_revenue
        FLOAT64 net_revenue
        INT64 total_item_quantity
    }
    fct_order_items {
        STRING order_item_key PK
        STRING order_key FK
        STRING item_key FK
        INT64 quantity
        FLOAT64 item_revenue_in_usd
    }
    fct_daily_funnel {
        STRING daily_funnel_key PK
        DATE date_key FK
        STRING channel_key FK
        INT64 sessions
        INT64 purchasing_sessions
        FLOAT64 revenue
    }
    fct_funnel_by_dim {
        STRING funnel_by_dim_key PK
        DATE date_key FK
    }

```

```

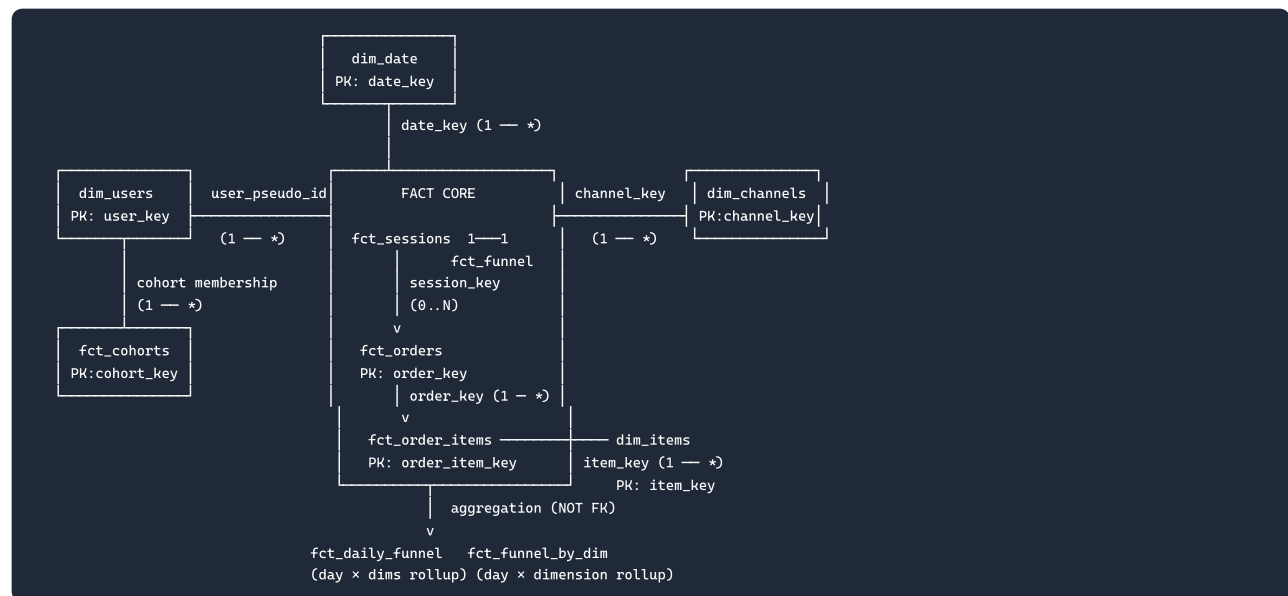
    STRING dimension
    STRING dimension_value
  }
  fct_cohorts {
    STRING cohort_key PK
    DATE cohort_week
    INT64 age_days
    INT64 cohort_size
    INT64 retained_users
  }

dim_date ||-o{ fct_sessions : "date_key"
dim_date ||-o{ fct_funnel : "date_key"
dim_date ||-o{ fct_orders : "date_key"
dim_date ||-o{ fct_daily_funnel : "date_key"
dim_date ||-o{ fct_funnel_by_dim : "date_key"
dim_users ||-o{ fct_sessions : "user_pseudo_id"
dim_users ||-o{ fct_orders : "user_pseudo_id"
dim_users ||-o{ fct_cohorts : "cohort membership"
dim_channels ||-o{ fct_sessions : "channel_key"
dim_channels ||-o{ fct_funnel : "channel_key"
dim_channels ||-o{ fct_orders : "channel_key"
dim_channels ||-o{ fct_daily_funnel : "channel_key"
dim_items ||-o{ fct_order_items : "item_key"
fct_sessions ||-| fct_funnel : "session_key (funnel EXTENDS session)"
fct_sessions ||-o{ fct_orders : "session_key (0..N orders)"
fct_orders ||-o{ fct_order_items : "order_key"

```

Aggregation (NOT a foreign key): `fct_funnel` is rolled up — COUNT / SUM of additive measures — into `fct_daily_funnel` (by day × channel × device × country × is\_new\_user) and into `fct_funnel_by_dim` (by day × canonical dimension). These rollups carry no row-level FK back to `fct_funnel`; they are pre-aggregated derivatives.

## 2.2 ASCII star schema



The facts sit in the center; the four conformed dimensions ( `dim_date` , `dim_users` , `dim_channels` , `dim_items` ) surround them. The two pre-aggregated rollups hang below the core as derived aggregations of `fct_funnel` , not as joined facts.

## 2.3 Relationship walk-through

- **dim\_date 1 — \* facts ( date\_key )**. Every event-grained and daily fact ( `fct_sessions` , `fct_funnel` , `fct_orders` , `fct_daily_funnel` , `fct_funnel_by_dim` ) carries one `date_key` pointing at exactly one calendar day; a day has many rows in each fact. This is the conformed time axis for all rollups.
- **dim\_users 1 — \* { fct\_sessions , fct\_orders } ( user\_pseudo\_id / user\_key )**; **dim\_users 1 — \* fct\_cohorts** . A user has many sessions and zero-or-more orders; cohort membership is a rollup of `dim_users` first-touch into weekly acquisition cohorts. Identity is cookie-grain, so a "user" here is a device+browser cookie, not a person.

- **dim\_channels 1** — \* { **fct\_sessions**, **fct\_funnel**, **fct\_orders**, **fct\_daily\_funnel** } ( **channel\_key** ). One of 10 channel groups attaches to many sessions/orders. Channel is session-scoped first-touch (the **traffic\_source** gotcha): the durable user-level **traffic\_source** is *first-touch* attribution, so the session-scoped **event\_params** source/medium is preferred and the user struct is only a fallback.
- **dim\_items 1** — \* **fct\_order\_items** ( **item\_key** ). One catalog product appears on many order lines; the items dimension never joins above the line-item grain.
- **fct\_sessions 1** — 1 **fct\_funnel** ( **session\_key** ). The funnel *extends* the session: exactly one funnel row per session row, same **session\_key**. **fct\_funnel** is the primary semantic grain; **fct\_sessions** adds engagement. Splitting them keeps each fact single-purpose while preserving a clean 1:1 join.
- **fct\_sessions 1** — 0..N **fct\_orders** ( **session\_key** ). A session yields zero or more orders — almost always 0 (most sessions don't buy) or 1. This is the bridge that lets revenue-per-session join order revenue back to the session universe.
- **fct\_orders 1** — \* **fct\_order\_items** ( **order\_key** ). One deduped order header explodes into one row per item line. Order-level totals live on the header; line-level revenue lives on the items.
- **Aggregation rollups (NOT FKs)**. **fct\_funnel** is aggregated into **fct\_daily\_funnel** (additive daily counts + revenue, the feed for Monitor/Decompose and the eval injector) and **fct\_funnel\_by\_dim** (funnel counts by one canonical dimension, the mix-vs-rate decomposition input). These are GROUP BY derivatives, carrying their own surrogate PKs and a **date\_key** FK, but no row-level foreign key back to **fct\_funnel**.

### 3. Master Table Catalog

Every table in the Helios pipeline, from raw source through the growth marts. Use these exact names, PKs, FKs, grains, and owners everywhere; the ER diagram above and every per-model section that follows must agree with this catalog.

| Table                        | Layer                         | Grain                   | Primary Key                                                                                                                    | Foreign Keys                                                                                                                     | Owner / Steward                   | Why it exists                                                                                                        |
|------------------------------|-------------------------------|-------------------------|--------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|----------------------------------------------------------------------------------------------------------------------|
| <b>src_ga4.events</b>        | raw                           | one row per EVENT       | none enforced (natural: <b>user_pseudo_id</b> , <b>event_timestamp</b> , <b>event_name</b> , <b>event_bundle_sequence_id</b> ) | none                                                                                                                             | Google / GA4 export               | Immutable; <b>event_params</b> only via <b>event_params</b> ; except this                                            |
| <b>stg_ga4__events</b>       | staging (view)                | EVENT                   | surrogate <b>event_id</b> or natural ( <b>user_pseudo_id</b> , <b>event_timestamp</b> , <b>event_name</b> )                    | → <b>src_ga4.events</b>                                                                                                          | analytics-eng / platform          | 1:1 typed, <b>session_id</b> , <b>page_location</b> , <b>device_id</b> , <b>purchase_id</b> downstream               |
| <b>stg_ga4__event_params</b> | staging (view)                | EVENT × PARAM KEY       | (event natural key, <b>param_key</b> )                                                                                         | → <b>src_ga4.events</b>                                                                                                          | analytics-eng / platform          | Flattens <b>event_params</b> once so n                                                                               |
| <b>int_ga4__sessionized</b>  | intermediate (ephemeral/view) | SESSION                 | <b>session_key</b>                                                                                                             | built from staging                                                                                                               | analytics-eng                     | KEYSTON never ship ( <b>user_pseudo_id</b> , landing, source/medium fallback), <b>engaged</b> , <b>ga_session_id</b> |
| <b>int_ga4__funnel_steps</b> | intermediate (ephemeral/view) | SESSION                 | <b>session_key</b>                                                                                                             | → <b>int_ga4__sessionized</b>                                                                                                    | analytics-eng                     | KEYSTON monotonic <b>session_id</b>                                                                                  |
| <b>fct_sessions</b>          | mart / core (table)           | SESSION (1 row/session) | <b>session_key</b>                                                                                                             | <b>user_pseudo_id</b> → <b>dim_users.user_key</b> ; <b>date_key</b> → <b>dim_date</b> ; <b>channel_key</b> → <b>dim_channels</b> | analytics-eng / Product Analytics | The config for engagement                                                                                            |

| Table                          | Layer                  | Grain                                                        | Primary Key                                                                                       | Foreign Keys                                                                                                                                                                                                                | Owner / Steward                             | Why it exists                                                              |
|--------------------------------|------------------------|--------------------------------------------------------------|---------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|----------------------------------------------------------------------------|
| <code>fct_funnel</code>        | mart / core (table)    | SESSION (1 row/session)                                      | <code>session_key</code> (also FK → <code>fct_sessions</code> )                                   | → <code>dim_users</code> ,<br><code>dim_date</code> ,<br><code>dim_channels</code>                                                                                                                                          | analytics-eng / Product Analytics           | The PRIMARY queries (wide dimension) here.                                 |
| <code>fct_daily_funnel</code>  | mart / growth (table)  | DAY × [channel_group, device_category, country, is_new_user] | <code>daily_funnel_key</code> (md5 of grain)                                                      | <code>date_key</code> → <code>dim_date</code> ;<br><code>channel_key</code> → <code>dim_channels</code>                                                                                                                     | analytics-eng / Growth Analytics            | Additive; revenue; 1 Decomposition Rates NO layer).                        |
| <code>fct_orders</code>        | mart / finance (table) | TRANSACTION (1 row/transaction_id)                           | <code>order_key</code> (= <code>transaction_id</code> )                                           | <code>session_key</code> → <code>fct_sessions</code> ;<br><code>user_pseudo_id</code> → <code>dim_users</code> ;<br><code>date_key</code> → <code>dim_date</code> ;<br><code>channel_key</code> → <code>dim_channels</code> | analytics-eng / Finance & Revenue Analytics | The deduplicate entity-of-refund/share revenue &                           |
| <code>fct_order_items</code>   | mart / finance (table) | TRANSACTION × ITEM LINE (1 row/order-item)                   | <code>order_item_key</code> (md5 of transaction_id + item_id + row)                               | <code>order_key</code> → <code>fct_orders</code> ;<br><code>item_key</code> → <code>dim_items</code>                                                                                                                        | analytics-eng / Finance & Revenue Analytics | The exploded revenue, i                                                    |
| <code>fct_funnel_by_dim</code> | mart / growth (table)  | DAY × DIMENSION                                              | composite ( <code>date_key</code> ,<br><code>dimension</code> ,<br><code>dimension_value</code> ) | <code>date_key</code> → <code>dim_date</code>                                                                                                                                                                               | analytics-eng / Growth Analytics            | Funnel room mix-vs-ra                                                      |
| <code>fct_cohorts</code>       | mart / growth (table)  | COHORT_WEEK × AGE_DAYS                                       | composite ( <code>cohort_week</code> ,<br><code>age_days</code> [, dims])                         | <code>cohort_week</code> derived from <code>dim_users</code> first-touch                                                                                                                                                    | analytics-eng / Growth Analytics            | Weekly active retained day_1/7, <code>fct_cohort</code> window ri cohorts. |
| <code>dim_users</code>         | mart / core (table)    | USER (1 row/user_pseudo_id)                                  | <code>user_key</code> (= <code>user_pseudo_id</code> )                                            | <code>first_channel_key</code> → <code>dim_channels</code>                                                                                                                                                                  | analytics-eng (conformed)                   | The USER touch attribute is_purcha                                         |
| <code>dim_items</code>         | mart / core (table)    | ITEM (1 row/item_id)                                         | <code>item_key</code> (= <code>item_id</code> )                                                   | none                                                                                                                                                                                                                        | analytics-eng (conformed)                   | The conformed (item_name) current pr                                       |
| <code>dim_channels</code>      | mart / core (table)    | CHANNEL_GROUP (10 rows)                                      | <code>channel_key</code>                                                                          | none                                                                                                                                                                                                                        | analytics-eng (conformed)                   | Conformed default grain is_paid/is                                         |
| <code>dim_date</code>          | mart / core (table)    | DAY (1 row/day)                                              | <code>date_key</code>                                                                             | none                                                                                                                                                                                                                        | analytics-eng (conformed)                   | Conformed 31; week/month for time-c                                        |

## Grain discipline

Every table in the catalog declares **exactly one grain**, and that grain is its contract. The four irreducible grains are:

- **SESSION** — one analytical visit, `session_key`. Carried by `int_ga4_sessionized`, `int_ga4_funnel_steps`, `fct_sessions`, and `fct_funnel`. `fct_funnel` is the canonical session grain the semantic layer resolves `sessions`, `users`, conversion rates, and `revenue` against.
- **TRANSACTION** — one completed order, `order_key` = `transaction_id`. Carried by `fct_orders` and exploded one level down to TRANSACTION × ITEM LINE in `fct_order_items`.
- **ITEM LINE** — one product line within an order, `order_item_key`; the only grain at which product-level revenue and attach rate are valid.

- **COHORT × AGE** — one weekly acquisition cohort at a given `age_days`, `cohort_key`; the grain for retention curves.

Plus the conformed dimension grains (DAY, USER, CHANNEL\_GROUP, ITEM) and the two pre-aggregated rollup grains ( `fct_daily_funnel` at day × dims; `fct_funnel_by_dim` at day × one dimension).

**Why mixing grains is forbidden.** A metric is only additive and reconcilable at its declared grain. Summing `session_revenue` at order grain, or counting `orders` at session grain, double-counts or under-counts because the row multiplicity differs across facts. The semantic layer enforces this: each metric is bound to one grain ( `revenue` → `fct_funnel`, `gross_revenue / aov` → `fct_orders`, retention → `fct_cohorts` ), and cross-grain comparisons are only sanctioned at the *grand total* (e.g. session-grain `revenue` reconciles to order-grain `gross_revenue` to within 0.5% in aggregate, but can legitimately differ per segment because the joins differ). Declaring one grain per table — and never widening or mixing it — is what makes every governed metric re-aggregate correctly across any slice the Decompose agent requests, and is the precondition for the "0 hallucinated columns / 100% governed SQL" target.

## Ownership model

**analytics-eng owns the pipeline end-to-end** — it builds and maintains every model from staging through the marts, owns all four conformed dimensions, and is accountable for the source-to-mart contracts. On top of that engineering ownership, each business domain assigns a **steward** who owns the *semantics* of a fact family: what the numbers mean, which caveats apply, and sign-off on changes that alter business meaning.

| Steward                     | Stewards which tables                                                                                  | Domain accountability                                                                                                                                 |
|-----------------------------|--------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Product Analytics           | <code>fct_sessions</code> , <code>fct_funnel</code>                                                    | Session/funnel semantics: engagement definition, monotonic <code>reached_*</code> flags, conversion-rate definitions.                                 |
| Finance & Revenue Analytics | <code>fct_orders</code> , <code>fct_order_items</code>                                                 | Revenue semantics: dedup-by-transaction, USD-only money, the gross/net/refund/shipping/tax split, AOV basis.                                          |
| Growth Analytics            | <code>fct_daily_funnel</code> ,<br><code>fct_funnel_by_dim</code> , <code>fct_cohorts</code>           | Aggregation and cohort semantics: additive-counts-only rule, mix-vs-rate decomposition inputs, retention denominators and the right-censoring caveat. |
| analytics-eng (conformed)   | <code>dim_date</code> , <code>dim_users</code> ,<br><code>dim_channels</code> , <code>dim_items</code> | Conformed dimension definitions: the 10 channel groups, the date spine, cookie-grain user identity, the product catalog.                              |

**What ownership means in practice.** (1) *Schema changes* — adding, renaming, or retyping a column requires the owning steward's review, because downstream the semantic layer and the eval labels bind to physical names; a rename is a breaking change. (2) *Tests* — the owner is responsible for the dbt tests on their tables: grain-integrity ( `unique` on the PK), `not_null` on keys, accepted-range on measures, funnel monotonicity, and revenue reconciliation. Keystone transforms ( `int_ga4__sessionized`, `int_ga4__funnel_steps`, revenue dedup) get golden-value tests because they fail silently. (3) *SLAs* — owners commit to freshness (daily; available ≤36h after `event_date` for the semantic-facing facts) and to the CI eval gate: no change to an owned model may drop top-1 diagnosis accuracy by more than the agreed threshold or introduce a hallucinated column. Conformed dimensions, owned solely by analytics-eng, carry the strictest change control because a change there ripples through every fact that joins them.